Supply Chain Late Delivery Risk Predictor

An End-to-End Data Science & AI Project

Fernando Gonzalez Navarro | Data Science & ML | 2026

Machine Learning

Gen AI

Business Intelligence

Project Overview

1. Business Problem
2. Data Strategy
3. Exploratory Analysis
4. Machine Learning Models
5. Gen AI Component
6. Business Impact & Demo

The Problem

55%

of orders flagged as
Late Delivery Risk

Customer Churn

Primary driver in e-commerce

Operational Costs

Reactive management is expensive

No Foresight

No system to predict delays proactively

Supply chain managers react to delays instead of preventing them.

The Solution: A 3-Layer Intelligence System

01 ANALYZE

Tableau Dashboard

Visualizes delivery bottlenecks by region and shipping mode

02 PREDICT

XGBoost ML Model

Identifies at-risk orders before they ship

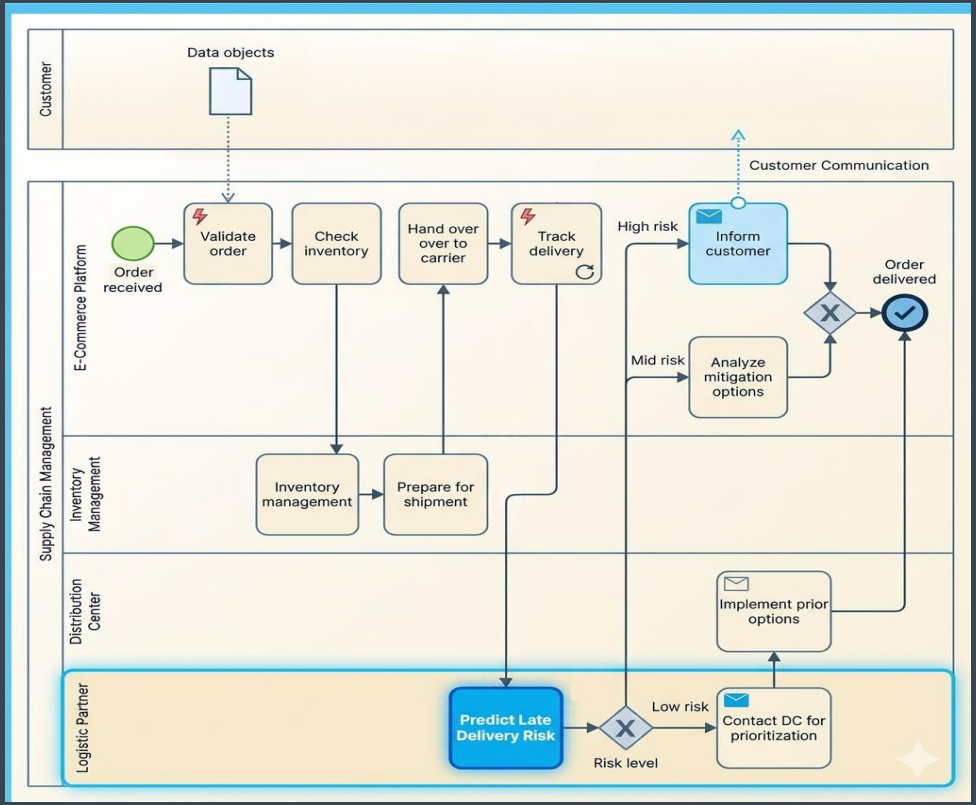
91% Accuracy
0.96 ROC-AUC

03 ACT

Gen AI - GPT-4o-mini

Generates proactive recovery actions and customer communications

The Process


Customer
Data objects
Customer Communication
Supply Chain Management
E-Commerce Platform
Order received
Validate order
Check inventory
Hand over over to carrier
Track delivery
High risk
Inform customer
Order delivered
Mid risk
Analyze mitigation options
Inventory Management
Inventory management
Prepare for shipment
Distribution Center
Implement prior options
Logistic Partner
Predict Late Delivery Risk
Risk level
Low risk
Contact DC for prioritization

Data Strategy

180,519
orders

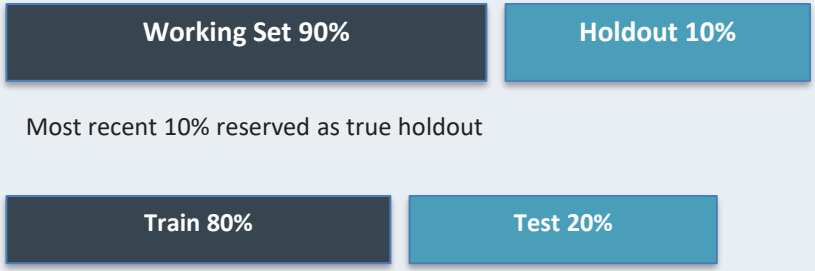
53
features

Jan 2015 → Jan 2018
time range

DataCo SMART SUPPLY CHAIN

source (Kaggle)

Holdout Strategy



Most recent 10% reserved as true holdout

Temporal split ensures holdout = future data

The Data Behind the Predictions

Excluded: Post-Delivery Data

- Actual days to deliver
- Final delivery outcome
- Order completion status
- Revenue earned per order

Excluded: Not Predictive

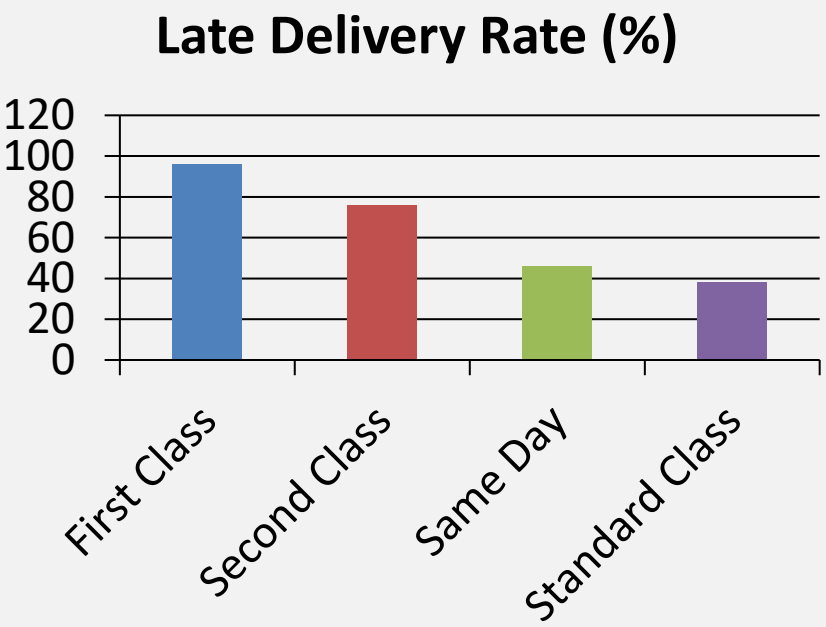
- Customer personal details
- Product images & descriptions
- Redundant location fields
- Duplicate identifier columns

Added Business Context

- Order timing: month, quarter, hour
- Product category (grouped)
- Shipping mode (binary signals)

Result: 37 clean, business-relevant signals | Zero missing data points

Key EDA Findings



The problem is NOT shipping speed.

First Class has the HIGHEST late rate (96%) because it overpromises tight 1-day delivery windows.

Orders scheduled for 1 day → 95% late rate
Orders scheduled for 4 days → 38% late rate

Root cause: Overpromising delivery windows, not execution failure.

Operational Intelligence: Tableau Dashboard

Late Delivery Rate by Shipping Mode

Late Delivery Rate by Country

Order Volume Over Time

Department/Category Performance

Interactive dashboard published on Tableau Public

Choosing the Right Approach

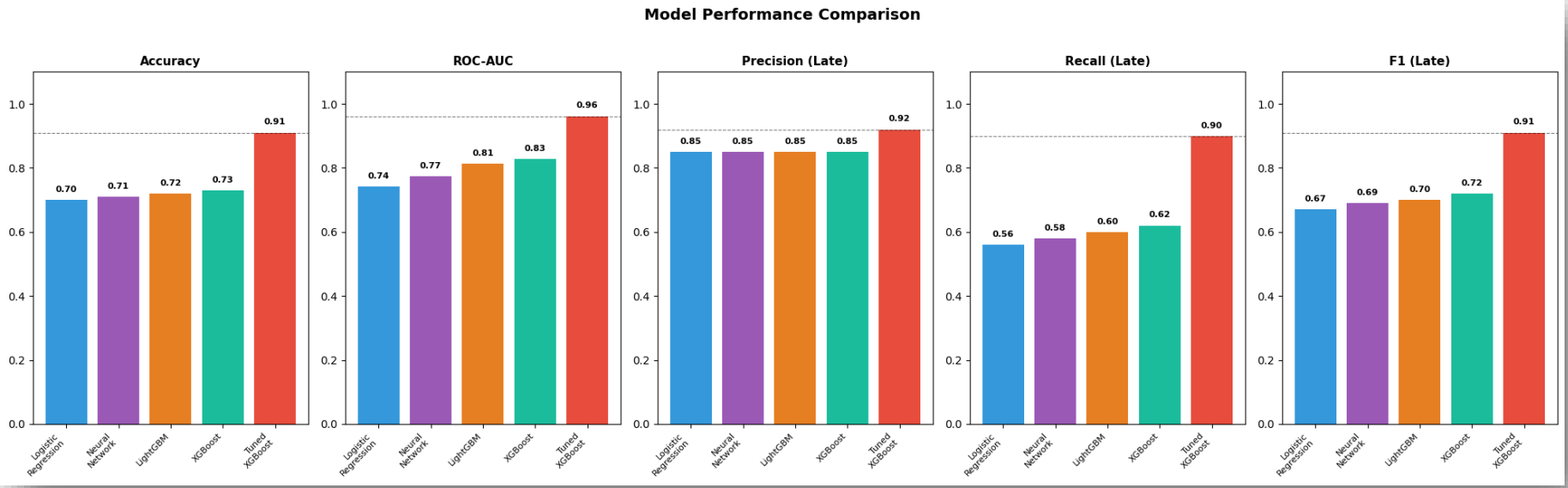
5 algorithms tested — one clear winner

Approach	Accuracy	Ranking Quality	Late Orders Caught	Balance Score	Missed Deliveries
Logistic Regression	70%	0.74	0.56	0.67	7,908
Neural Network	71%	0.77	0.58	0.69	7,446
LightGBM	72%	0.81	0.60	0.70	7,120
XGBoost	73%	0.83	0.62	0.72	6,748
Tuned XGBoost	**91%**	**0.96**	**0.90**	**0.91**	**1,743**

Advanced tuning cut missed at-risk orders by 77% — from 7,908 down to 1,743

Choosing the Right Approach

5 algorithms tested — one clear winner



Model Results — In Business Terms

91%
Orders correctly classified

0.96
Prediction quality index

On-Time Orders Cleared
12,693

Low-Risk Orders Over-Flagged
1,998

90%
Late deliveries caught pre-ship

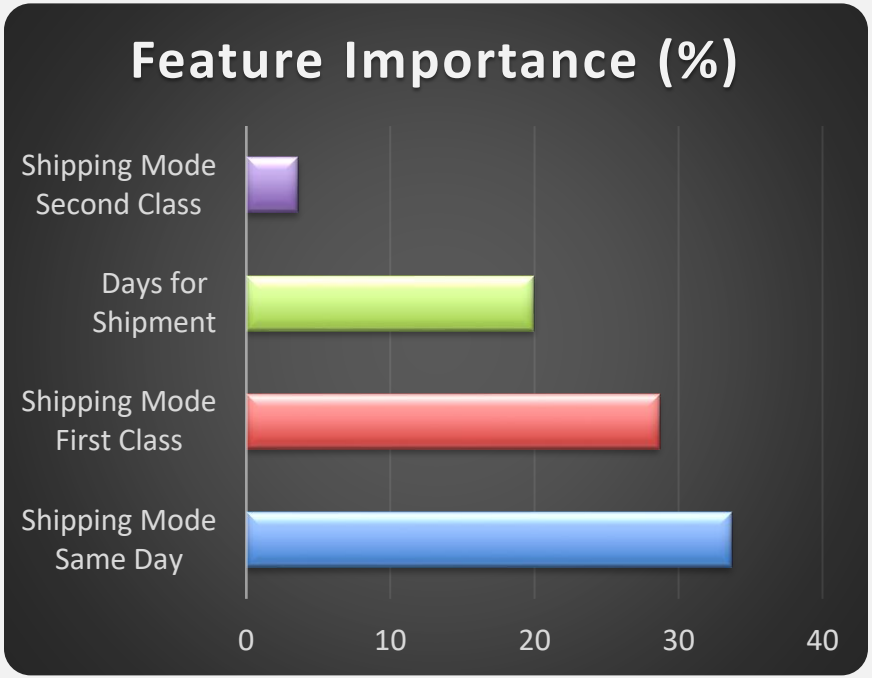
1,743
At-risk orders missed

At-Risk Orders Missed
1,743

Late Orders Caught Early
16,060

Trained on 129,973 orders | Validated on 32,494 orders

What Drives Late Delivery Risk?



Shipping mode alone explains 66% of predictive power

Scheduled delivery window is the 3rd most important signal

Geographic and product features play minor roles

The model learned what EDA showed: the commitment matters more than the capability.

Validated on Genuinely New Data

Training & Test Period

Accuracy: 91%
Data range: 2015–2017

Business Patterns Evolved

Future Unseen Data

Accuracy: 69%
Data range: Aug 2017–Jan 2018

The performance difference is expected — not a flaw. Business patterns naturally change over time: new products, seasonal shifts, carrier behaviours. This is why regular model retraining is recommended in production.

Understanding the Accuracy Gap

91% accuracy on historical data → 69% on future data — a business signal, not a model failure

1 Q4 & Peak Season Pressure

The holdout covers Aug 2017 – Jan 2018: the most volatile period in global e-commerce. Black Friday, Cyber Monday, and the holiday rush create volume spikes and carrier strain that behave nothing like the 2015–2017 patterns the model was trained on.

2 New Products with No Delivery History

As the catalog evolved, new SKUs entered with zero historical delivery data. The model had never seen their shipping profiles, lead times, or warehouse handling requirements, so it defaulted to patterns that no longer applied.

3 Carrier Network Disruptions

Logistics providers adjust routes, add seasonal partners, and reprioritize capacity during peak season. These operational shifts weren't reflected anywhere in 2015–2017 training data, making carrier-related predictions less reliable.

4 A Static Model in a Dynamic Market

Once deployed, a model cannot learn from its own outcomes. Without retraining, every passing month widens the gap between what the model "knows" and how the business actually operates. This is concept drift — and it's normal.

Takeaway: This gap is a roadmap. It tells us exactly what comes next — more recent training data, real-time carrier signals, and a feedback loop that keeps the model current with the business.

Proactive Recovery: Gen AI Pipeline

High-Risk Order Detected

↓

Model Context Extracted

↓

GPT-4o-mini Prompt

↓

Recovery Action Generated

Example: First Class | South Asia | 100% Risk

Customer Message:
Dear Valued Customer, we sincerely apologize for the potential delay...

Action 1: Contact carrier to expedite processing

Action 2: Review nearest distribution center inventory

Upgrade: Switch from First Class to Express Shipping

Product: Streamlit Web Application

App Features

- Input form: Shipping mode, region, product, customer segment
- Real-time ML prediction with probability score
- AI recovery action for high-risk orders
- Deployed on Streamlit Community Cloud

Live Demo

supply-chain-risk-predictor.streamlit.app

Tech Stack

Python | XGBoost | OpenAI GPT-4o-mini | Streamlit

Business Impact

77%

Reduction in false negatives after tuning (7,908 → 1,743)

9,691

High-risk orders flagged in holdout period for proactive action

< 1s

Prediction + recovery action generation per order

Each false negative represents an order that will be late without proactive intervention. At scale, identifying 77% more at-risk orders before dispatch translates directly to reduced customer churn and lower expediting costs.

What Prediction Adoption Unlocks

From a single model to a smarter supply chain — the compounding value of getting started

TODAY

Reactive Becomes Proactive

- Catch 90% of late orders before they leave the warehouse
- Automated Gen AI recovery actions in under 1 second
- Proactive customer alerts reduce complaint volume
- Supply chain team acts on data, not gut feel

NEXT 6–12 MONTHS

Fix the Root Cause, Not Just the Symptom

- Redesign delivery window commitments by shipping mode
- Build carrier performance scoring into procurement decisions
- Differentiate SLAs by product category and region
- The model revealed WHAT to fix — now the business can act on it

FUTURE DATA STRATEGY

A Self-Improving Intelligence System

- Collect real-time carrier capacity and weather feeds
- Add customer satisfaction scores as a model input
- Create a retraining pipeline driven by live outcomes
- Each passing month makes the model — and the process — smarter

Key insight: The data this model couldn't fully explain is exactly the data worth collecting next. Prediction doesn't just add value — it reveals where process improvement and smarter data collection should go.

Key Learnings & Challenges

Technical Learnings

- Data leakage is the silent killer
- Temporal splits are more honest than random splits
- Tree models outperform neural networks on tabular data
- Hyperparameter tuning (Optuna) had highest ROI

Business Learnings

- Root cause (overpromising) was hidden in the data
- 69% holdout accuracy more valuable than 91% same-period
- Gen AI quality depends on structured context
- Concept drift is production reality, not failure

Conclusion

This project demonstrates a complete intelligence pipeline — from raw supply chain data to a deployed predictive application with AI-powered decision support. The system identifies late delivery risk before orders ship, enabling proactive intervention rather than reactive management.

91%
Test Accuracy

0.96
ROC-AUC

Live
App

Next Steps

- Regular model retraining to address concept drift
- Ensemble methods for marginal improvement
- Integration with real-time ERP systems

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